

Assignment #2

Question 1

For a 2-terminal MOS structure with p-type body and gate material work function greater than the silicon work function ($\phi_M < \phi_S$); draw the energy band diagrams for flat band conditions: (a) assuming zero oxide trapped charge and (b) an oxide trap charge of Q'_0 . Also, draw the energy band diagrams in (c) accumulation, and (d) depletion & inversion conditions assuming zero oxide trapped charge.

Question 2

Calculate the flat band voltage for an n-type substrate with $N_D = 5 \times 10^{17} \text{ cm}^{-3}$, an SiO_2 insulator with $t_{\text{ox}} = 2 \text{ nm}$, and polysilicon gate doped n-type with $N_D = 10^{20} \text{ cm}^{-3}$. Assume $Q'_0 = 2 \times 10^{-9} \text{ C/cm}^2$, $kT/q = 25.9 \text{ mV}$ at $T = 300 \text{ K}$ and intrinsic carrier concentration $n_i = 1.18 \times 10^{10} \text{ cm}^{-3}$.

Question 3

Show that the channel charge in a MOS system is given by

$$Q'_C = \sqrt{2q\epsilon_{\text{Si}}N_A} \sqrt{\phi_t e^{-\psi_s/\phi_t} + \psi_s - \phi_t + e^{-2\phi_F/\phi_t} (\phi_t e^{\psi_s/\phi_t} - \psi_s - \phi_t)}$$

Question 4

Prove that the channel charge per unit area can be expressed by the following expression in a two terminal MOS structure-

$$Q'_c = -\text{sgn}(\psi_s) \sqrt{2q\epsilon_S N_A} \sqrt{\psi_s + \phi_t e^{(\psi_s - 2\phi_F)/\phi_t}}$$

Question 5

What is ψ_{sa} ? Show that its value is given by

$$\psi_{sa} = \left(-\gamma/2 + \sqrt{\gamma^2/4 + V_{GB} - V_{FB}} \right)^2$$

Question 6

What is meant by small signal capacitance? Show that depletion region incremental capacitance C'_b per unit area is given by

$$C'_i = \sqrt{2q\epsilon_{si}N_a} \frac{1}{2\sqrt{\psi_s + \phi_t} e^{(\psi_s - 2\phi_F)/\phi_t}}$$

Question 7

Show that inversion region incremental capacitance C'_i per unit area is given by

$$C'_i = \sqrt{2q\epsilon_{si}N_a} \frac{e^{(\psi_s - 2\phi_F)/\phi_t}}{2\sqrt{\psi_s + \phi_t} e^{(\psi_s - 2\phi_F)/\phi_t}}$$

Question 8

Show that in depletion (with ψ_s larger than a few ϕ_t) and in weak inversion C'_{gb} is approximately given by

$$C'_{gb} = \frac{C'_{ox}}{\sqrt{1 + \frac{4}{\gamma^2} (V_{GB} - V_{FB})}}$$